

Mon

Name

M#

Solve the following ODEs

■ $y' = (1+y)e^t$ $y(0) = 1$

$$\ln(1+1) = e^0 + C$$

$$C = \ln(2) - 1$$

sep

$$dy/dt = (1+y)e^t$$

$$\int \frac{dy}{1+y} = \int e^t dt + C$$

$$\ln(1+y) = e^t + C$$

1. $y' = \frac{\sin(y)}{\cos(y)} (t^2 + 1) = dy/dt$

$$\int \frac{\cos(y)}{\sin(y)} dy = \int (t^2 + 1) dt + C$$

$$u = \sin(y)$$

$$du = \cos(y)$$

$$\ln(|\sin(y)|) = \frac{t^3}{3} + t + C$$

$$\blacksquare t^2 y' + t y = t^3 \text{ with } y(1) = 2$$

$$t y' + y = t^2$$

$$(t y)' = t^2$$

$$t y = t^3/3 + C$$

$$y = t^2/3 + C/t$$

$$2. e^t y' + 2 e^t y = e^t \text{ with } y(0) = 1$$

$$y' + 2y = 1$$

$$v(t) = e^{2t}$$

$$e^{2t} y' + 2 e^{2t} y = e^{2t}$$

$$(e^{2t} y)' = e^{2t}$$

$$e^{2t} y = \int e^{2t} + C$$

$$e^{2t} y = \frac{1}{2} e^{2t} + C$$

or

$$y = \frac{1}{2} + C e^{-2t}$$

lin

$$y' + 1/t y = t$$

$$y' + p(t)y = s(t)$$

$$v(t) = e^{\int 1/t dt}$$

$$v(t) = e^{\ln(t)} = t$$

$$C = 2 - 1/3$$

$$2 = 1/3 + C/1$$